

Lab T4 - Personal Report

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1. Problem

Camera-LiDAR timestamp offset creates position misalignment in ADAS fusion. LiDAR captures at t_{ref} minus offset while fusion uses the ego pose at t_{ref} . This is a synthetic 2-D alignment benchmark.

2. Method

TV3 owns ego trajectory, timestamp convention, noise and synthetic ground truth. Scenario C uses a constant-turn-rate trajectory with $R=30$ m and speed 10 m/s.

3. Benchmark

The group ran 45 conditions and 4,500 samples. All 0 ms baselines are 0.023-0.027 m. A at 100 ms and $d=20$ m gives $E_{pre}=2.001$ m and linear $E_{post}=0.026$ m.

4. Failure case

C at $d=40$ m and $offset=100$ ms has linear $E_{post}=1.351$ m, above the group-defined 0.5 m threshold. CTRV gives 0.025 m on the same samples.

5. Engineering decision

Use CTRV with a reliable offset and yaw rate. If offset exceeds 30 ms and yaw rate exceeds 0.2 rad/s, avoid LiDAR-to-camera-bbox association, track independently, and log a synchronization warning.

Evidence: results/results.csv, results/samples.csv, results/run_log.txt, plots/error_vs_offset.png, plots/alignment_turn.png. Command: `python -m src.run_benchmark --config config_chot.yaml`.